System Information Commands

DEVOPS-002 · System Information · Sprint 9 (29/6 - 12/7)

These commands gather information about a Linux machine's kernel, hardware resources, storage, and network configuration - useful first checks on any new or unfamiliar system.

System & Kernel Information

Command	What It Displays
`uname -a`	Shows kernel name, hostname, kernel version, architecture, and OS, all in a single line.
`hostnamectl`	Displays the hostname along with OS, kernel version, architecture, and virtualization details.
`uptime`	Shows how long the system has been running, the number of logged-in users, and the system load average.

Hardware & Resources

Command	What It Displays
`lscpu`	Lists CPU details: number of cores/threads, model name, architecture, and clock speed.
`lsblk`	Lists all block storage devices (disks, partitions) and their mount points, shown as a tree.
`free -h`	Shows RAM and swap usage (total / used / free) in human-readable units such as MB or GB.
`df -h`	Shows disk space usage for each mounted filesystem, in human-readable units.

Network Information

Command	What It Displays
`ip addr`	Shows network interfaces and their assigned IP addresses - the current, standard tool.
`ifconfig`	Shows network interfaces and IP addresses - an older, legacy tool with the same basic purpose.

Note: ip addr and **ifconfig** do the same basic job. **ifconfig** comes from the older **net-tools** package and is not installed by default on many newer Linux distributions, while **ip addr** is the current standard, part of the **iproute2** package.